

Belkasoft X

Report on a Digital Forensics and Mobile Forensics Platform

1. Introduction

Belkasoft X, formerly known as Belkasoft Evidence Center X, is a commercial digital-forensics and incident-response platform developed by Belkasoft. It is designed as an end-to-end forensic environment: investigators can acquire evidence, process and recover artifacts, analyze the evidence, correlate information, and prepare reports from a single application.

The platform supports several branches of digital forensics, including computer forensics, mobile forensics, memory/RAM forensics, and cloud forensics. For mobile investigations, its main targets are Android and iOS devices. This makes Belkasoft X relevant when an investigation involves smartphones as well as computers or other digital sources.

2. What Belkasoft X Actually Does

At a practical level, Belkasoft X performs four major functions:

- **Acquisition:** obtains forensic data from supported devices, backups, images, cloud sources, and other evidence sources.
- **Parsing and recovery:** processes raw evidence and identifies structured forensic artifacts such as messages, calls, browser records, files, media, and application databases.
- **Analysis and correlation:** lets investigators search, filter, visualize timelines and locations, and correlate artifacts across multiple devices and sources.
- **Reporting:** allows relevant findings to be bookmarked, exported, and assembled into an investigation report.

Belkasoft X is therefore not merely a file viewer. It combines evidence acquisition with automated artifact extraction and forensic examination. The platform supports a large range of digital artifact types.

3. Working Architecture

The working architecture can be understood as a pipeline in which different acquisition and analysis components feed a common case environment.

Layer	Function
Evidence sources	Android/iOS phones, computer storage, disk images, RAM dumps, cloud sources, backups and other supported forensic images.
Acquisition layer	Uses the appropriate acquisition method for the source, such as ADB, iTunes backup, agent-based acquisition, checkm8, rooted-device extraction, chipset-specific Android methods, or imported forensic images.

Layer	Function
Processing / parsing layer	Reads the acquired data, recognizes file systems and application structures, parses databases and application artifacts, and recovers relevant information.
Case / correlation layer	Stores and organizes evidence for searching, filtering, bookmarking, timeline analysis, geolocation and cross-device correlation.
Analysis interface	Provides high-level artifact views and lower-level viewers such as file-system, hexadecimal, SQLite, Registry and plist viewers.
Reporting layer	Exports selected findings and produces investigation reports containing relevant artifacts and contextual information.

The architecture is not tied to one extraction technique. The acquisition layer selects a method appropriate to the device and available access. The resulting data then enters a common examination environment where artifacts from different sources can be searched and correlated. Belkasoft X provides both high-level artifact analysis and low-level examination through file-system, hexadecimal, SQLite, Registry and plist viewers.

4. Mobile Forensics: Android

For Android, Belkasoft X provides multiple acquisition approaches because Android devices differ by manufacturer, chipset, security configuration, Android version, and access level. Standard approaches include ADB backup and agent-based acquisition. Rooted devices can provide logical or physical filesystem acquisition. Belkasoft also supports specialized approaches such as Qualcomm EDL, MediaTek/MTK acquisition, Spreadtrum acquisition, APK downgrade, and automated screen capture.

A simplified Android workflow is:

Android device → identify supported acquisition method → acquire backup/image/data → parse filesystem and application data → extract artifacts → analyze and correlate → report.

The acquisition method determines how much information can be obtained. An ordinary ADB backup is fundamentally different from a physical or filesystem extraction from a rooted device. Therefore, the examiner must select the most appropriate method available rather than assuming that every method produces the same evidence.

5. Mobile Forensics: iOS

For iOS, Belkasoft X supports several acquisition mechanisms. These include iTunes backup acquisition, lockdown-file-based acquisition, AFC, agent-based acquisition, checkm8-based acquisition for compatible devices, acquisition from jailbroken devices, crash-log extraction, and screen capture. The exact method available depends on the device model, iOS version, lock state, and other technical conditions.

For example, an iTunes backup is a relatively standard acquisition route, whereas checkm8-based acquisition can provide a fuller extraction on compatible older Apple devices. Checkm8 applies to a defined range of Apple hardware and is not a method for every modern iPhone.

6. Types of Evidence Analyzed

After acquisition, Belkasoft X can organize and present a wide range of digital artifacts. Examples include:

- Calls and SMS
 - Instant-messaging and social-media application data
 - Contacts and email
 - Photographs, videos and audio
 - Browser history and other web artifacts
 - Documents and application databases
 - GPS and other geolocation information
 - Cryptocurrency-related artifacts
 - Fitness and device activity data
 - System and application files
 - Cloud-related data
 - RAM and other computer-forensic artifacts

The platform also provides tools for relationship analysis and map-based visualization of supported geolocation artifacts.

7. Analysis Architecture

Belkasoft X provides both high-level and low-level analysis. At the high level, the investigator sees interpreted artifacts such as chats, emails, images, calls and browser records. At the low level, the investigator can inspect the underlying filesystem and database structures using dedicated viewers. This is important because automated parsing provides a convenient interpretation, while low-level examination allows the investigator to validate the underlying evidence.

The general relationship is:

Raw evidence → filesystem/database structures → artifact parsers → interpreted artifacts → search/correlation → investigator validation → report.

8. Typical Working Procedure

1. **Identify the evidence source:** determine the phone, backup, forensic image, cloud source, or other data source to be examined.
2. **Select an acquisition method:** choose a method appropriate to the device and its access state.
3. **Acquire the data:** create the forensic extraction or import an existing forensic image/backup.
4. **Process the evidence:** allow Belkasoft X to parse supported filesystems, databases and application artifacts.
5. **Examine artifacts:** review messages, calls, media, locations, browser activity and other relevant evidence.
6. **Search and correlate:** use keywords, filters, timelines, maps and connection analysis to connect events and sources.
7. **Validate important findings:** inspect the underlying files or databases when necessary instead of relying solely on an automatically parsed artifact.
8. **Report:** bookmark relevant evidence and generate a report containing the findings and their context.

9. Advantages

- **End-to-end workflow:** acquisition, examination, analysis and reporting are available in one platform.
- **Multiple mobile acquisition methods:** Android and iOS require different techniques, and Belkasoft provides several alternatives.
- **Broad artifact coverage:** automated parsing reduces the amount of manual examination required.
- **Cross-device correlation:** multiple datasets can be examined within the same case and connected through relationship and communication analysis.
- **High- and low-level examination:** investigators can move from interpreted artifacts to underlying files and databases when validation is necessary.
- **Supports more than phones:** the same platform can handle computer, memory, cloud and other supported evidence sources.

10. Limitations

Belkasoft X is powerful, but it does not remove the technical limitations of the devices being investigated.

- **Acquisition is device-dependent:** available extraction methods vary by model, chipset, operating-system version, lock state and security configuration. Not every method works on every phone.
- **Some advanced methods are restricted to particular devices:** for example, checkm8 is applicable to a defined range of Apple hardware rather than modern iPhones in general.
- **Acquisition can be intrusive:** methods involving rooting, jailbreaking, exploits, agents or application downgrade may have greater forensic or operational implications than standard backups.
- **Encryption and security controls remain significant:** application encryption, device encryption and secure hardware can limit what can be obtained.
- **Automated parsing is not infallible:** investigators must understand the underlying artifact and validate important conclusions.
- **Commercial licensing:** Belkasoft X is proprietary software, so access to the full platform depends on the applicable license/edition.
- **Large evidence sets require resources:** processing many devices and large datasets requires sufficient storage, memory and processing capacity.

11. Belkasoft X in Mobile Forensics

For mobile-phone investigations, Belkasoft X can be viewed as a chain of three major stages:

Stage	What happens
Acquisition	Data is extracted from Android/iOS using a method supported by the device and its security state.
Processing and examination	The acquired image, backup or filesystem is parsed into recognizable forensic artifacts.
Analysis and reporting	The investigator searches, correlates, validates and documents the relevant findings.

This architecture is important because acquisition and analysis are separate technical problems. A forensic examiner may have excellent analysis software but still obtain limited evidence if the acquisition method only provides a restricted backup. Conversely, a comprehensive extraction is useful only if it can be correctly parsed and interpreted.

12. Conclusion

Belkasoft X is a commercial digital-forensics platform that provides an end-to-end environment for acquiring, processing, examining, correlating and reporting digital evidence. For mobile forensics, its major strength is the range of acquisition methods available for Android and iOS devices, followed by automated parsing of application and system artifacts.

Its working architecture can be summarized as evidence source → acquisition method → forensic image/backup → processing and artifact parsing → analysis and correlation → validation → reporting. The platform's limitations are mainly determined by the security and technical characteristics of the source device. A forensic tool cannot guarantee complete extraction from every phone, and advanced acquisition methods may be restricted to particular models or circumstances.

Therefore, Belkasoft X should be understood not simply as a phone data recovery tool, but as a complete forensic examination platform in which the quality and completeness of the final evidence depend on both the acquisition technique and the investigator's subsequent analysis and validation.

13. References

1. Belkasoft — Mobile Forensics Software and Belkasoft X product documentation.
2. Belkasoft X Help — Android acquisition documentation.
3. Belkasoft X Help — iOS acquisition documentation.
4. Belkasoft X Help — Evidence analysis and artifact documentation.